COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S2
DATE PERFORMED:	08/05/2026	**DATE SUBMITTED:**	
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Hands-on Activity 5.1 Queues

1. PROCEDURE OUTPUT

What procedures did you perform?

```
//ILOA
#include <iostream>
#include <queue>

void display(std::queue<char> copyQ);

int main() {
    //create an object:
    std::queue<char> myQ;
    //use the enqueue operation
    myQ.push('K');
    myQ.push('H');
    myQ.push('E');
    std::cout<<"Current back is: "<<myQ.front()<<std::endl;
    std::cout<<"Current back is: "<<myQ.back()<<std::endl;
    display(myQ);

    //use the dequeue operation
    myQ.pop();
    display(myQ);
    myQ.pop();
    display(myQ);
    myQ.pop();

    //check the queue if it is empty
    std::cout<<"is the queue empty?"<<myQ.empty() <<std::endl;
    return 0;
}
```

```
//note only use the number of the member functions of thue queue STL
void display(std::queue<char> copyQ){
    //create a copy of the queue
    std::queue<char> temp = copyQ;

    //loop until empty
    while(!temp.empty()) {
        //display the front
        std::cout<< " " <<temp.front()<< "";
        //dequeue the front
        temp.pop();
    }
    //add a new line
    std::cout << std::endl;
}
```

```
//ILOB
#include <iostream>
#include "queueLL.h"

int main() {
    qNode<char>* front = nullptr;
    qNode<char>* back = nullptr;

    enqueue('K', &front, &back);
    std::cout << "front: " << front->data << " " << "back: " <<
back->data << std::endl;

    enqueue('H', &front, &back);
    std::cout << "front: " << front->data << " " << "back: " <<
back->data << std::endl;

    enqueue('E', &front, &back);
    std::cout << "front: " << front->data << " " << "back: " <<
back->data << std::endl;
```

```
    std::cout << "testing the dequeue operator: \n";

    dequeue(&front, &back);
    std::cout << "front: " << front->data << " " << "back: " <<
back->data << std::endl;

    dequeue(&front, &back);
    std::cout << "front: " << front->data << " " << "back: " <<
back->data << std::endl;

    dequeue(&front, &back);
    dequeue(&front, &back);

    return 0;
}
```

```
#ifndef QUEUELL_H
#define QUEUELL_H

//create a class node
template <typename T>
class qNode{
    public:
        T data;
        qNode* next;
};

//new node creator
template <typename T>
qNode<T>* new_node(T newData){
    //allocate a space for the new node
    qNode<T>* newNode = new qNode<T>;

    //store the newdata to the newNode
    newNode ->data = newData;

    //point the newNode next to null
    newNode ->next = nullptr;
```

```
    return newNode;
}

//enqueue
template <typename T>
void enqueue(T newData, qNode<T>** frontPtr, qNode<T>** backPtr) {
    //create a new node
    qNode<T>* newNode = new_node(newData);
    //check if the queue is empty
    if((*frontPtr)==nullptr && (*backPtr) == nullptr){
        (*frontPtr) = newNode;
        (*backPtr)= newNode;
        return;
    }
    //if not then insert at the back
    (*backPtr)->next = newNode;
    (*backPtr) = newNode;
}

//dequeue
template <typename T>
void dequeue(qNode<T>** frontPtr, qNode<T>** backPtr) {
    //check if the queue is empty
    if((*frontPtr) == nullptr && (*backPtr) == nullptr){
        std::cout<<"the queue is empty "<<std::endl;
        return;
    }
    //create a temporary variable to store the node to be deleted
    qNode<T>* deleteTemp = nullptr;
    //assign the current front to the deleteTemp
    deleteTemp = (*frontPtr);
    //check if the queue has only 1 item
    if ((*frontPtr)->next == nullptr && (*backPtr)->next == nullptr) {
        // means only 1 item inside the queue
        (*frontPtr) = nullptr;
        (*backPtr) = nullptr;
        delete deleteTemp;
```

```
        return;
    }

    (*frontPtr) = (*frontPtr)->next;
    deleteTemp->next = nullptr;
    // delete node
    delete deleteTemp;
}

//display all
template <typename T>
void display(qNode<T>* frontPtr) {

}

//isEmpty

#endif
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

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Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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